Developmental Foundations of Tangible Interactive Learning Systems for Early Childhood:

The Böðn Project

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April 18, 2026

Abstract

Böðn is an open-source, battery-powered tangible learning system designed for children aged 3–7. The device combines physical controls (buttons, rotary encoders, toggle switches) with multimodal feedback (LED animations, synthesized audio, screen graphics) to deliver play experiences that target core executive functions—working memory, inhibitory control, and cognitive flexibility—as well as sensorimotor, cognitive, language, and social-emotional development. This paper reviews the developmental science that informs Böðn's design, maps each feature to established developmental domains, and identifies areas for future development. We argue that tangible interfaces offer advantages over touchscreen-only systems for early childhood learning, and that game-based, guided play can meaningfully support executive function development during the critical preschool window.

Keywords: executive functions, tangible user interfaces, early childhood, guided play, physical computing, open-source hardware

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1 Introduction

The preschool years (ages 3–7) represent a critical period for the development of executive functions (EF)—the cognitive control processes that enable goal-directed behaviour (Diamond, 2013). A large body of evidence links early EF to school readiness (Blair and Razza, 2007), social competence, and lifelong well-being (Moffitt et al., 2011). Unlike IQ, which is relatively stable, executive functions are highly malleable during early childhood and respond to targeted practice (Diamond, 2013; Montroy et al., 2016).

At the same time, the proliferation of touchscreen devices has raised concerns about passive screen time in young children (Radesky et al., 2015). The World Health Organization recommends no more than one hour of sedentary screen time per day for children aged 2–4 (World Health Organization, 2019). This creates a tension: digital systems can deliver adaptive, engaging practice that physical toys cannot, yet the medium itself may undermine the developmental benefits.

Tangible user interfaces (TUI)—physical objects that bridge the digital and physical worlds—offer a potential resolution. Since Ishii and Ullmer (1997) introduced the concept, a growing literature has explored TUI for educational contexts, finding advantages in engagement, spatial understanding, and embodied learning (Marshall, 2007; Price et al., 2003).

This paper presents **Böðn**, an open-source tangible learning system that combines physical controls with adaptive, game-based experiences designed around the developmental science of early childhood. We review the theoretical foundations, describe the system design, map each feature to developmental domains, and identify directions for future work.

2 Background

2.1 Executive Functions in Early Childhood

Miyake et al. (2000) proposed the influential three-factor model of executive functions: *working memory* (updating), *inhibitory control* (inhibition), and *cognitive flexibility* (shifting). While originally validated in adults, subsequent work has confirmed this structure in preschoolers, with important developmental nuances (Montroy et al., 2016).

Working memory is the ability to hold and manipulate information in mind. In 4-year-olds, typical capacity is 2–4 items (Diamond, 2013). Tasks like remembering a sequence of coloured lights and reproducing it from memory directly exercise this capacity.

Inhibitory control encompasses both response inhibition (suppressing a prepotent motor response) and interference control (resisting distracting information). The classic "day–night" task, where children must say "night" when shown a sun card, illustrates the

challenge for preschoolers (Diamond, 2013).

Cognitive flexibility requires disengaging from one rule set and engaging a new one. The Dimensional Change Card Sort (DCCS) task (Zelazo, 2006) is the standard assessment: children sort cards first by colour, then by shape. Most 3-year-olds fail the switch; most 5-year-olds succeed. The transition period (ages 3–5) represents a critical window for practice.

2.2 Guided Play and Playful Learning

Weisberg et al. (2013) define guided play as adult-scaffolded activities where children explore within a structured environment designed to make discovery productive. Guided play occupies a middle ground between free play (child-directed, no adult structure) and direct instruction (adult-directed, no child agency). Evidence suggests that guided play produces deeper learning than either extreme for concept acquisition and problem-solving (Hirsh-Pasek et al., 2015).

2.3 Tangible User Interfaces for Young Children

Ishii and Ullmer (1997) introduced the concept of tangible user interfaces as a way to "give physical form to digital information." For young children, who are still developing fine motor skills and abstract thinking, the physical affordances of tangible interfaces offer several advantages over touchscreens:

- **Embodied interaction**: Physical manipulation engages motor planning, proprioception, and spatial reasoning simultaneously (Price et al., 2003).
- **Reduced cognitive load**: Physical affordances (a button looks pressable, a knob looks turnable) reduce the interface learning burden, freeing cognitive resources for content (Marshall, 2007).
- **Multimodal feedback**: Tactile confirmation (button click, encoder detent) combines with visual and auditory feedback to strengthen sensorimotor associations.

2.4 Multimodal Feedback and Sensory Integration

Young children benefit from redundant, multimodal feedback—receiving the same information through multiple sensory channels simultaneously (Price et al., 2003). In Böðn, every user action produces visual (LED/screen), auditory (tone/sound), and tactile (physical click/detent) feedback. This redundancy:

- Strengthens cross-modal neural connections
- Provides confirmation even if one channel is missed
- Supports different learning preferences and sensory profiles

2.5 Auditory Processing and Musical Exposure

Early exposure to structured sound environments supports auditory processing skills that underpin language acquisition and musical development (Kraus and Chandrasekaran, 2010; Trainor, 2005). Active sound production (as opposed to passive listening) is particularly effective.

2.6 Bilingualism and Executive Function

A substantial literature documents enhanced executive functions in bilingual children, attributed to the constant need to manage two active language systems (Bialystok, 2011). The bilingual advantage is most pronounced for inhibitory control and cognitive flexibility—the same skills that Böðn's game modes target.

The bilingual advantage has been contested: Paap et al. (2015) argued that positive results reflect publication bias rather than a robust effect. However, in the specific context of early childhood (ages 3–5), where language systems are still being established, the evidence for enhanced inhibitory control remains more consistent (Bialystok, 2011). Sweden provides a particularly favourable setting for early bilingualism: English-language media exposure is pervasive, and Swedish preschools increasingly introduce English informally.

Letter Case and Early Literacy

Research on letter-name knowledge shows that preschoolers recognise uppercase letters before lowercase ones—uppercase knowledge predicts lowercase acquisition with a facilitation factor exceeding 16× (Piasta, 2014). However, since more than 95% of printed text uses lowercase forms, prolonged exposure to only uppercase delays the transition to reading real text (Treiman and Kessler, 2006). Böðn therefore uses **title case** (first letter uppercase, remainder lowercase) for all child-facing labels, giving the child the familiar uppercase anchor while building lowercase familiarity.

Dual-Language Labeling

Byers-Heinlein and Lew-Williams (2013) demonstrated that bilingual infants distinguish their two languages from birth using rhythmic and phonological cues, countering the common concern that simultaneous exposure causes confusion. Tan and Byers-Heinlein (2024) showed that *translation equivalents*—knowing a word in one language—significantly increases the probability of learning the corresponding word in the other language, especially at younger ages. Classroom research on dual-language labeling confirms that presenting objects labelled in both languages supports vocabulary development in both (Castro et al., 2011). Böðn's printed NFC cards therefore present both Swedish and English labels simultaneously (e.g. "Katt / Cat"), while the on-device UI shows one language at a time

(switchable in settings) to respect the display's space constraints.

2.7 Pretend Play and Social-Cognitive Development

Lillard et al. (2013) reviewed the evidence linking pretend play to cognitive development, finding associations with theory of mind, narrative skills, and emotional understanding. While the causal direction remains debated, the correlational evidence is strong, and pretend play is universally observed across cultures during the preschool years.

2.8 Feedback, Failure, and Growth Mindset

Dweck (2006) distinguishes between fixed mindset (ability is innate) and growth mindset (ability develops through effort). The type of feedback children receive after failure shapes which mindset they develop. Process-focused feedback ("You tried a different strategy, good!") promotes growth mindset; outcome-focused feedback ("Wrong!") promotes fixed mindset.

Böðn's feedback design follows this principle: errors produce soft, warm cues (gentle tone, dim red glow) rather than punitive signals. The intent is to frame failure as information, not judgement.

3 System Design

3.1 Hardware Architecture

Böðn is built around an ESP32-S3 microcontroller with the following physical interface components:

- 8 mini push buttons (colour-coded, via MCP23017 I2C expander)
- 5 arcade buttons with PWM-controlled LEDs (via PCA9685)
- 2 toggle switches (via MCP23017)
- 3 rotary encoders with push buttons (navigation, volume, throttle)
- 2.8" ILI9341 TFT primary display (240×320, landscape)
- 1.8" ST7735 TFT secondary display (128×160, portrait)
- 2× WS2812 8-LED sticks + 92-LED perimeter strip (RGB)
- MAX98357A I2S amplifier with 3W speaker
- INMP441 I2S MEMS microphone
- 6600 mAh LiPo battery with thermal monitoring (DS18B20×2)

3.2 Software Architecture

The firmware is written in MicroPython and follows a modular architecture:

- **Rule engines** (`*_rules.py`): Pure logic modules with no hardware dependencies, testable via standard pytest on the host.
- **Screen modules** (`ui/*.py`): Render game state to displays and LEDs; handle input events.
- **Audio engine**: 16-voice native C mixer running on a dedicated CPU core, with priority channels (music, SFX pool, UI feedback) and built-in tone generators (sine, square, sawtooth, noise).
- **Session manager**: Enforces play time limits, breaks, and quiet hours.
- **i18n system**: All user-facing strings localised (Swedish default, English available).

4 Developmental Analysis

This section maps each Böðn feature to established developmental domains and analyses coverage. Table 1 presents the full feature–domain matrix.

4.1 Feature–Domain Mapping

4.1.1 Simon (Härmapa)

The Simon memory game directly targets **working memory** capacity. The starting sequence length of 2 items is calibrated to the lower bound of 4-year-old WM capacity (Diamond, 2013). Each successful round adds one item, providing adaptive difficulty within the child's zone of proximal development (Vygotsky, 1978).

4.1.2 Rule Follow (Regelleken)

Modelled after the DCCS task (Zelazo, 2006), Rule Follow alternates between "match" and "opposite" rules. The "opposite" rule requires **inhibitory control** (suppressing the prepotent matching response), while the rule switch exercises **cognitive flexibility**. The frequency of rule switches (every 4–6 correct answers) is within the range used in DCCS research.

4.1.3 Mystery Box (Mysterium)

An open-ended **cause-and-effect** exploration mode with no failure state. The design follows the "guided play" model (Weisberg et al., 2013): the environment is structured (specific button–colour mappings, discoverable combinations) but the child's exploration is entirely self-directed.

4.1.4 Flöde

A **spatial reasoning** and **problem-solving** puzzle that requires aligning gaps in vertical walls. The snap-to-grid mechanic reduces motor precision demands while preserving the cognitive challenge.

4.1.5 Garden of Life (Trädgården)

A **pattern recognition** mode based on Conway's Game of Life. The child plants coloured seeds and observes emergent behaviour—an implicit introduction to complex systems and self-organisation. The mode exercises no executive functions but provides rich visual stimulation and cause-and-effect learning.

4.1.6 Soundboard (Ljudbräda)

An **auditory processing** and **open exploration** mode. Customisable sound banks allow parents to create themed experiences. The volume encoder provides fine motor practice with immediate auditory feedback.

4.1.7 Spaceship Cockpit (Rymdskepp)

The most developmentally comprehensive mode, targeting **cognitive flexibility** (5 scenario types requiring different inputs), **listening comprehension** (TTS-driven instructions), and **imaginative play** (structured pretend-play as a spaceship captain). The adaptive difficulty system adjusts timeout windows based on the child's recent performance, maintaining an appropriate challenge level.

4.1.8 Story Mode (Saga)

A data-driven branching narrative engine where the child listens to narrated scenes and makes choices via arcade buttons to steer the plot. Primary targets: **listening comprehension** (must listen to narration to choose), **bilingual exposure** (full narration in both Swedish and English), and **imaginative play** (guiding a character through a narrative world). Stories are defined as script files on the SD card; adding new stories requires no code changes. The first story adapts Beatrix Potter's *The Tale of Peter Rabbit* (public domain) into a choose-your-own-adventure with four endings.

4.1.9 High-Five Friends (Heja-kompisar)

A reaction-time game targeting **hand-eye coordination** and **inhibitory control**. Animals "pop up" on illuminated arcade buttons and the child must tap the lit button before it disappears. The reaction window shrinks from 2 s to 0.5 s as the child succeeds, providing adaptive difficulty within the child's zone of proximal development (Vygotsky, 1978).

Gentle miss feedback (soft sound, dim animation) and automatic restart after three misses support **persistence** while keeping emotional stakes low (Dweck, 2006). The mode also exercises **auditory processing** (distinct hit/miss sounds) and **self-regulation** (managing the frustration of near-misses).

4.1.10 Sortera — NFC Classification Game

A tangible implementation of the Dimensional Change Card Sort (DCCS) task (Zelazo, 2006), targeting **cognitive flexibility** and **inhibitory control**. The device announces a sorting rule ("Find the animals!"), and the child scans NFC cards. After several correct matches the rule switches to a different dimension ("Find all the red ones!"), requiring the child to inhibit the previous sorting strategy. Sortera is the first mode to provide **active bilingual reinforcement**: on each correct scan, the screen displays both the Swedish and English word (e.g. "Katt / Cat") matching the physical card, and bilingual TTS announces both words—primary language first, then secondary (Tan and Byers-Heinlein, 2024). This dual-label approach leverages the translation-equivalent bootstrapping effect while keeping game instructions in the configured language for clarity.

4.1.11 Räkna — Progressive Math with Tangible Numbers

A six-level NFC math mode targeting **pattern recognition** (subitising), **sequential thinking** (counting and equation order), **problem solving** (composing equations step by step), and **fine motor** coordination (handling multiple cards per equation). Räkna implements Bruner's Concrete–Pictorial–Abstract progression (Bruner, 1966; Willingham, 2017): a physical dot-pattern card is placed on the reader (concrete), the screen renders matching dots (pictorial), and from Level 4 onwards the numeral and operator cards are introduced (abstract). Dice-face dot patterns at low levels rather than numerals reflect the developmental evidence that subitising is the foundation of number sense, not numeral recognition (Clements and Sarama, 2014). Levels span free exploration (ages 3–4), quantity matching and more/less comparison (4–5), addition and subtraction with tokens (5–6), and symbolic equations (6–7+). A left-to-right linear number path on the display implements Siegler and Ramani's (2009) finding that board-game-style number lines measurably improve preschool number competence. Bilingual TTS announces each quantity in Swedish and English, carrying the translation-equivalent bootstrapping effect (Tan and Byers-Heinlein, 2024) into a cognitively engaging math task. The mode is always self-paced and never timed — a deliberate design choice given the evidence that timed math exercises produce anxiety even in preschoolers. Together with Sortera, Räkna addresses the early-math domain that Duncan et al. (2007) identified as the single strongest predictor of later academic achievement.

4.1.12 Blippa — Free-Play NFC Discovery

A goal-less, failure-less companion to the other NFC modes targeting **open exploration**, **cause and effect**, and **imaginative play**. Tapping any owned card—from any card set the device knows about— produces a full-screen emoji, a bilingual label, and either a procedural POS "blip" or an optional per-card sample. There is no rule to follow, no score, and no wrong scan: the mode exists specifically to turn a growing collection of physical cards into a low-friction exploration toy. This design is motivated by evidence that children's causal learning is strongest when exploration is self-directed rather than instructed (Bonawitz et al., 2011), and it intentionally creates a surface for the cashier and transit-gate pretend play that 3–5-year-olds gravitate to, linking open exploration with imaginative play. Unknown card identifiers within a known card set produce a deterministic "mystery blip" whose pitch is derived from a stable hash of the id, so a given card always sounds the same—reinforcing object permanence and rehearsal of cause-effect contingencies.

4.1.13 Sequencer (Sequenser)

A loop-based step sequencer targeting **working memory**, **sequential thinking**, **pattern recognition**, and **creative expression**. Five percussion tracks (arcade buttons) and one melody track (eight mini buttons mapping to a pentatonic scale) are recorded into an 8- or 16-step grid during looped playback. Toggle switches control play/pause, step count, and metronome—making abstract musical concepts tangible. The mode introduces computational thinking through creative play (Papert, 1980): building patterns, running them, hearing something unexpected, and debugging by toggling individual steps. BPM is adjustable via encoder (70–140), providing fine motor practice with immediate auditory feedback.

4.2 Coverage Matrix

4.3 Gap Analysis

The matrix reveals strong coverage for executive functions, visual processing, open exploration, and—with the addition of Sequencer and High-Five Friends—auditory processing, hand-eye coordination, and persistence. Story Mode significantly improved three previously identified gaps: listening comprehension, bilingual exposure, and imaginative play each now have two primary-level features (Spaceship + Story Mode). The Sequencer filled the creative expression gap, and High-Five Friends added a dedicated hand-eye coordination mode. The two NFC modes that followed—Sortera and Räkna—add a tangible manipulation layer and address two previously underserved domains: Sortera provides a tangible Dimensional Change Card Sort implementation (Zelazo, 2006) that strength-

Table 1: Feature–developmental domain coverage matrix. ● ● ● = primary target; ●● = significant; ● = incidental; blank = not applicable.

Domain	*Simon*	*Rule Follow*	*Mystery Box*	*Flöde*	*Garden*	*Soundboard*	*Spaceship*	*Story Mode*	*High-Five*	*Sequencer*	*Sortera*	*Bä*
Working memory	● ● ●	●●		●			●●	●●	●	● ● ●	●	●
Inhibitory control	●	● ● ●					●●		●●		●●	●
Cognitive flexibility		● ● ●					● ● ●			●●	● ● ●	●
Cause & effect			● ● ●		●●	●●		●●	●●	●●	●●	●
Pattern recognition	●●		●●		● ● ●					● ● ●	● ● ●	● ●
Spatial reasoning				● ● ●								●
Sequential thinking	● ● ●			●●				●●		● ● ●		● ●
Problem solving				● ● ●			●			●●		●
Fine motor	●	●	●	●●	●	●	●●	●	●	●●	●●	●
Hand–eye coord.	●●	●●		●●			●●		● ● ●	●●		●
Auditory processing	●●					● ● ●	●●	● ● ●	●●	●●	●●	●
Visual processing	●●	●●	● ● ●	●●	● ● ●		●●	●●	●●	●●	●●	● ●
Listening comp.						●	● ● ●	● ● ●			●	●
Bilingual exposure	●	●		●	●	●	●	● ● ●	●	●	●●	●
Self-regulation	●●	● ● ●			●		●●	●	●●		●●	●
Persistence	●●	●		● ● ●			●	●	●●	●●	●	●
Imaginative play							● ● ●	● ● ●		●●		
Open exploration			● ● ●		● ● ●	● ● ●					●	●
Divergent thinking			● ● ●		●●					● ● ●		●
Creative expression					●●	● ● ●				● ● ●		

ens cognitive flexibility and inhibitory control, while Räkna opens the early-mathematics domain that Duncan et al. (2007) identified as the single strongest predictor of later academic achievement. Remaining areas with limited coverage include:

1. **Spatial reasoning**: Still primarily Flöde; Räkna's linear number path contributes incidentally. A tangram or construction mode would add breadth.
2. **Problem solving**: Flöde (primary), Sequencer (debugging), and Räkna (equation building) now all contribute; a scavenger-hunt style NFC puzzle mode would further broaden the domain.
3. **Bilingual exposure**: Story Mode provides full narration in both languages, and Sortera and Räkna add active bilingual reinforcement (dual-language labels and TTS on each card scan). A dedicated language-switching game could further leverage the bilingual advantage for EF development (Bialystok, 2011).

5 Discussion

5.1 Tangible Interfaces vs. Touchscreen Applications

5.2 Adaptive Difficulty and the Zone of Proximal Development

5.3 Feedback Design and Growth Mindset

5.4 Session Limits and Healthy Technology Use

5.5 Limitations

6 Future Work

1. **NFC tangible interaction layer**: Integrate a PN532 NFC reader to enable tagged physical objects (cards, stickers, figurines) as input. Research on tangible user interfaces consistently shows that physical manipulation benefits learning in preschoolers through reduced cognitive load, embodied engagement, and support for classification tasks (Marshall, 2007; Antle, 2013; Manches and O'Malley, 2012). Planned NFC modes include:

    - *Sortera* — a tangible classification game implementing the DCCS task (Zelazo, 2006) with physical cards, targeting cognitive flexibility and inhibitory control.
    - *Saga Builder* — tangible storytelling where scanned character/setting cards trigger narration, extending the existing Story Mode with a richer physical input method (Sylla et al., 2012).

- *Vocabulary Explorer* — bilingual word cards for active language switching, leveraging the bilingual EF advantage (Bialystok, 2011).
- *Memory Match* — a physical search-and-select task that adds motor planning to working memory challenges (Manches and O'Malley, 2012).
- *Räkna* — a progressive math mode using dot-pattern NFC cards as tangible number tokens. Follows Bruner's Concrete-Pictorial-Abstract progression (Bruner, 1966): physical tokens (concrete) → dot displays (pictorial) → numerals (abstract). Six levels spanning subitising, counting, comparison, addition, and subtraction (ages 3–7). Grounded in evidence that early math is the strongest predictor of later academic achievement (Duncan et al., 2007), that linear number board games produce measurable and persistent gains in preschoolers' number competence (Siegler and Ramani, 2009), and that kindergarten number sense predicts math achievement through third grade (Jordan et al., 2009).

The NFC interaction pattern (find object → bring to reader → hold) gives children physical agency in a way that fixed buttons cannot, and has been commercially validated by products like Toniebox and Yoto Player for children as young as 3. See the companion document `nfc_tangible_learning.md` for the full research summary.

2. **Empirical validation**: Conduct pre–post studies comparing EF measures (DCCS, digit span) in children using Böðn vs. control activities.

3. **Record & Replay mode**: Implement voice recording and playback with effects (pitch shift, echo) to strengthen creative expression, auditory processing, and bilingual exposure.

4. **Cooperative play**: Add turn-taking and collaborative modes to support social-cognitive development.

5. **Adaptive difficulty modelling**: Implement Bayesian difficulty estimation to maintain each child in their zone of proximal development across play sessions.

6. **Longitudinal tracking**: Use the existing session logging infrastructure to track developmental progress over months and years.

7. **Active bilingual features**: Develop language-switching games that leverage the bilingual EF advantage.

7 Conclusion

Böðn demonstrates that tangible, open-source hardware can deliver developmentally informed play experiences for young children. By grounding each feature in established

developmental science—particularly the executive function literature—and by providing physical controls rather than touchscreen interaction, the system aims to combine the adaptability of digital systems with the developmental benefits of hands-on play. The open-source design invites collaboration from educators, developmental psychologists, and the maker community to extend and validate the approach.

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